

JEE MAIN REVISION MASTER: MAGNETIC EFFECT OF CURRENT SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 209 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Lorentz Force, Cyclotron & Magnetic Force on Wires	108 PYQs	Magnetic Forces
Very High	Biot-Savart Law & Ampere's Circuital Law	104 PYQs	Magnetic Field

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Lorentz Force, Cyclotron & Magnetic Force on Wires	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$, $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$, $T = \frac{2\pi m}{qB}$, $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$
Biot-Savart Law & Ampere's Circuital Law	$dB = \frac{\mu_0 I dl \sin\theta}{4\pi r^2}$, $B_{\text{wire}} = \frac{\mu_0 I}{2\pi d}$, $B_{\text{center}} = \frac{\mu_0 N I}{2R}$, $B_{\text{solenoid}} = \mu_0 n I$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Lorentz Force, Cyclotron & Magnetic Force on Wires (Very High Yield — 108 PYQs)

Core Principle: Trajectory of charged particles in magnetic fields, velocity selector ($\vec{E} \perp \vec{B}$), and force between parallel currents.

■■ **Caution / JEE Pitfall:** Magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$ does ZERO work because $\vec{F} \perp \vec{v}$. It changes direction of velocity, never kinetic energy!

■ **Speed Trick:** Parallel currents in the SAME direction ATTRACT; anti-parallel currents REPEL.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A current carrying circular loop of radius 2 cm with unit normal $\vec{n} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ is placed in a magnetic field, \$..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "An insulated wire is wound so that it forms a flat coil with $N=200$ turns. The radius of the innermost turn is $r_1=3\text{cm}$, and of the out..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "The magnetic field at the centre of a current carrying circular loop of radius R is $16\mu\text{ T}$. The magnetic field at a distance $x=\sqrt{...}$ " [Ans: B]

3.2 Biot-Savart Law & Ampere's Circuital Law (Very High Yield — 104 PYQs)

Core Principle: Calculation of magnetic fields generated by steady currents in wires, loops, and solenoids.

■ **Caution / JEE Pitfall:** In solenoid formula $B = \mu_0 n I$, n is number of turns per UNIT LENGTH ($n = N/L$), not total turns N .

■ **Speed Trick:** Right-hand thumb rule: thumb points along current, curled fingers give direction of circular magnetic field lines.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "A circular coil of radius 2 cm and 125 turns carries a current of 1 A . The coil is placed in a uniform magnetic field of magnitude 0.4 T" **[Ans: 314]**
- **[2026 Evening Shift] Q#2 (MCQ):** "A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm . The magne..." **[Ans: B]**
- **[2026 Morning Shift] Q#2 (MCQ):** "A current carrying solenoid is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along ..." **[Ans: D]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{3\hat{i} + 2\hat{j}}{\sqrt{13}}$ is placed in a magnetic field, $\vec{B} = B_0(3\hat{i} + 2\hat{j})$. If $B_0 = 4 \times 10^{-3} \text{ T}$ and current $I = 100\sqrt{2} \text{ A}$, the torque experienced by the loop is ____ Wb.A. ($\pi = 3.14$)

- (A) $16 \times 10^{-5} \text{ k}$
- (B) $5024 \times 10^{-7} \text{ k}$
- (C) $5024 \times 10^{-7} \text{ i}$
- (D) $5024 \times 10^{-7} \text{ j}$ **[Correct]**

Correct Answer: D

Solution Analysis: When a current-carrying loop is placed in an external magnetic field, it experiences a torque ($\vec{\tau}$) . The magnetic properties of the loop are represented by its magnetic dipole moment ($\vec{m}$) , defined as : $\vec{m} = I\vec{A}$ Where I is the current and $\vec{A}$ is the area vector of the loop. The direction of the area vector is perpendicular to the plane of the loop, given by the unit normal $\hat{n}$. $\vec{A} = A\hat{n}$ The torque is given by the cross product of the magnetic moment and the magnetic field : $\vec{\tau} = \vec{m} \times \vec{B} \Rightarrow \tau = IAB \sin\theta$

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

A circular coil of radius 2 cm and 125 turns carries a current of 1 A . The coil is placed in a uniform magnetic field of magnitude 0.4 T . The axis of the coil makes an angle of 30° with the direction of the magnetic field. The torque acting on the coil is $\alpha \times 10^{-4} \text{ N.m}$. The value of α is ____ . ($\pi = 3.14$)

Correct Answer: 314

Solution Analysis: The magnetic dipole moment of a single loop of wire carrying current I and enclosing area A is given by $m = IA$ For a coil consisting of N turns, the total magnetic moment is : $m = NIA$ Since the coil is circular with radius r , the area A is πr^2 . $m = NI(\pi r^2)$ The torque experienced by a magnetic dipole in a uniform magnetic field $\vec{B}$ is given by the vector cross product : $\vec{\tau} = \vec{m} \times \vec{B}$ The magnitude of this torque is : $\tau = mB \sin\theta$ Where θ is the angle between the magnetic moment (which is along the axis of the coil) and the magnetic field